DR. SARAH HENSLEY AI RAG — PRODUCTION RUNBOOK

ENVIRONMENT: PRODUCTION (AWS EC2)
HOST: agent.doctorsarahhensley.com
IP: 18.118.138.91 | **VERSION:** 2.2.0
STATUS: ACTIVE / HARDENED

1. System Architecture & Port Binding Inventory

All internal services are strictly isolated to loopback interface (127.0.0.1) and reverse-proxied through Nginx with HSTS, TLS 1.3, and strict security headers.

Service	Internal Binding	Runtime / Supervisor	Role & Isolation Policy
Nginx Proxy	0.0.0.0:80 / 443	systemd / nginx	TLS Termination, HSTS, Rate Limiting, Static Asset Cache
FastAPI Backend	127.0.0.1:8000	systemd / dsh-rag.service	RAG Orchestration, SSE Streaming, User Personalization
Qdrant Vector DB	127.0.0.1:6333	Native Qdrant daemon	384-dim MiniLM vector search on collection documents
MySQL 8.4 Server	127.0.0.1:3306	systemd / mysqld	Relational CRM customer profiles & conversation memory
SQLite DBs	Embedded (*.db)	In-Process WAL mode	users_data.db & ingestion_tracker.db local cache

2. Operational Management & Deployment Playbook

The application runs continuously as an unprivileged systemd service under the ubuntu user with automatic crash recovery (Restart=always, RestartSec=3s).

Core CLI Commands:

- Check Status: sudo systemctl status dsh-rag.service
- Restart Service: sudo systemctl restart dsh-rag.service
- Live App Logs: journalctl -u dsh-rag.service -f
- Live Chat Stream: tail -f /home/ubuntu/DSH-RAG/logs/chat.log
- Production Deploy: cd /home/ubuntu/DSH-RAG && git pull && sudo systemctl restart dsh-rag.service

3. API Endpoints & Zero-Trust Security Policy

All incoming traffic is governed by strict zero-trust identity enforcement. Unauthenticated public callers are mathematically locked to user_id='guest' to prevent IDOR user impersonation.

Endpoint	Method	Auth Requirement	Rate Limit	Purpose / Security Rule
/chat	POST	Public / Session Token	15 req/min	Interactive chat; forces unverified calls to 'guest'
/chat/stream	POST	Public / Session Token	15 req/min	Live SSE token stream with Layer 2 greeting guardrail
/health	GET	Public (Sanitized)	None	High-level readiness; redacts internal IPs & file paths
/documents	GET	Master X-API-Key	Admin	Protected; lists all 19 verified knowledge base files
/drive/status	GET	Master X-API-Key	Admin	Protected; returns Google Drive monitor connection status
/drive/sync	POST	Master X-API-Key	Admin	Protected; triggers immediate manual Google Drive re-scan
/api/v1/ask	POST	Master X-API-Key	Admin	Headless RAG retrieval for external service integrations
/api/v1/cache/clear	POST	Master X-API-Key	Admin	Flushes in-memory Semantic Cache and User Profile Cache

4. Token Optimization & Safeguard Architecture

The generation engine enforces a multi-layer safeguard system to guarantee cost control, prompt caching discounts, and zero mid-sentence response cutoffs.

Safeguard Layer	Mechanism	Target Threshold	Operational Impact
Layer 1: Static Prompt Cache	Byte-identical system prompts	100% Static Prefix	Unlocks ~50% API cost discount via provider prompt caching
Layer 2: Dynamic Token Ceilings	Lane-specific max_tokens	250 / 700 / 800 tok	Short: 250, Clinical: 700, Sales: 800 (Prevents 2500-token runaway)
Layer 3: Greeting Guardrail	Deterministic Python regex	Turn 1 / Turn 2+	Forces 'Hi {Name},' on Turn 1; strips repetitive greetings on Turn 2+
Layer 4: Truncation Trimmer	clean_trailing_truncation()	Terminal punctuation	Snaps output to last full sentence; eliminates dangling '**Practical'

5. Automated Schedules, Logrotate & Maintenance

Operational maintenance is fully automated via Linux crontab and logrotate to prevent disk exhaustion over prolonged high-volume production use.

Schedule (IST)	Script / Command	Target Assets	Retention & Outcome
Daily 02:00 AM	sync_wordpress_to_mysql.py	WooCommerce REST API	Syncs new orders, purchases, & quiz results to MySQL
Daily 03:00 AM	backup_databases.py --local-only	SQLite DBs & MySQL Dump	Zero-lock compressed snapshots; saves 98% space
Sunday 04:00 AM	run_maintenance.py	SQLite DBs & systemd journal	Runs VACUUM, updates query stats, vacuums 7d journal
Automated Hourly	/etc/logrotate.d/dsh-rag	All logs/*.log files	Rotates at 10MB; retains 7 compressed .gz files; zero downtime

6. Knowledge Base & Google Drive Ingestion Operations

Dr. Sarah Hensley's clinical library is continuously synchronized from Google Drive into Qdrant vectors.

Folder Configuration: Google Drive Folder ID: 13iyfLkaWOvvw8JQVyzjw_YjTec9cZ_PG
Service Account: /home/ubuntu/DSH-RAG/service_account.json (Encrypted credentials)
Current Status: 19 clinical documents ingested and marked READY in Qdrant.
Manual Re-Index Trigger: curl -X POST https://agent.doctorsarahhensley.com/drive/sync -H 'X-API-Key: dsh_rag_secret_key_2026'
Verification Endpoint: curl -s https://agent.doctorsarahhensley.com/documents -H 'X-API-Key: dsh_rag_secret_key_2026' | jq

7. Incident Response Playbook & Disaster Recovery

Follow these standard operating procedures during system alerts, server restarts, or data recovery operations.

Incident Scenario	Diagnostic Steps	Resolution Procedure
502 Bad Gateway / Process Crash	1. sudo systemctl status dsh-rag.service 2. journalctl -u dsh-rag.service -n 50	Check for port 8000 collisions or missing environment keys in .env. Execute sudo systemctl restart dsh-rag.service.
Qdrant Vector DB Offline	1. curl -s http://127.0.0.1:6333/healthz 2. ps aux | grep qdrant	Restart Qdrant service: sudo systemctl restart qdrant.service. Verify port 6333 responds on localhost.
Primary Groq LLM API Outage	Inspect logs/application.log for provider timeout errors.	Automatic failover is built-in: system automatically diverts to secondary OpenAI gpt-4o-mini. No manual restart required.
Database Recovery from Snapshot	Locate latest compressed snapshot in /home/ubuntu/DSH-RAG/backups/.	1. Stop service: sudo systemctl stop dsh-rag.service 2. Restore SQLite: gunzip -c backups/users_data_*.db.gz > users_data.db 3. Restore MySQL: gunzip -c backups/rag_users_*.sql.gz | mysql -u rag_user -p rag_users 4. Start service: sudo systemctl start dsh-rag.service

OPERATIONAL SIGN-OFF & CONTACT INFORMATION
- Primary Maintainer: DSH Engineering & Operations Team
- Production Server Host: AWS EC2 18.118.138.91 (us-east-2)
- Public Health Check: https://agent.doctorsarahhensley.com/health
- Automated Backup Directory: /home/ubuntu/DSH-RAG/backups/ (14-day rolling retention)